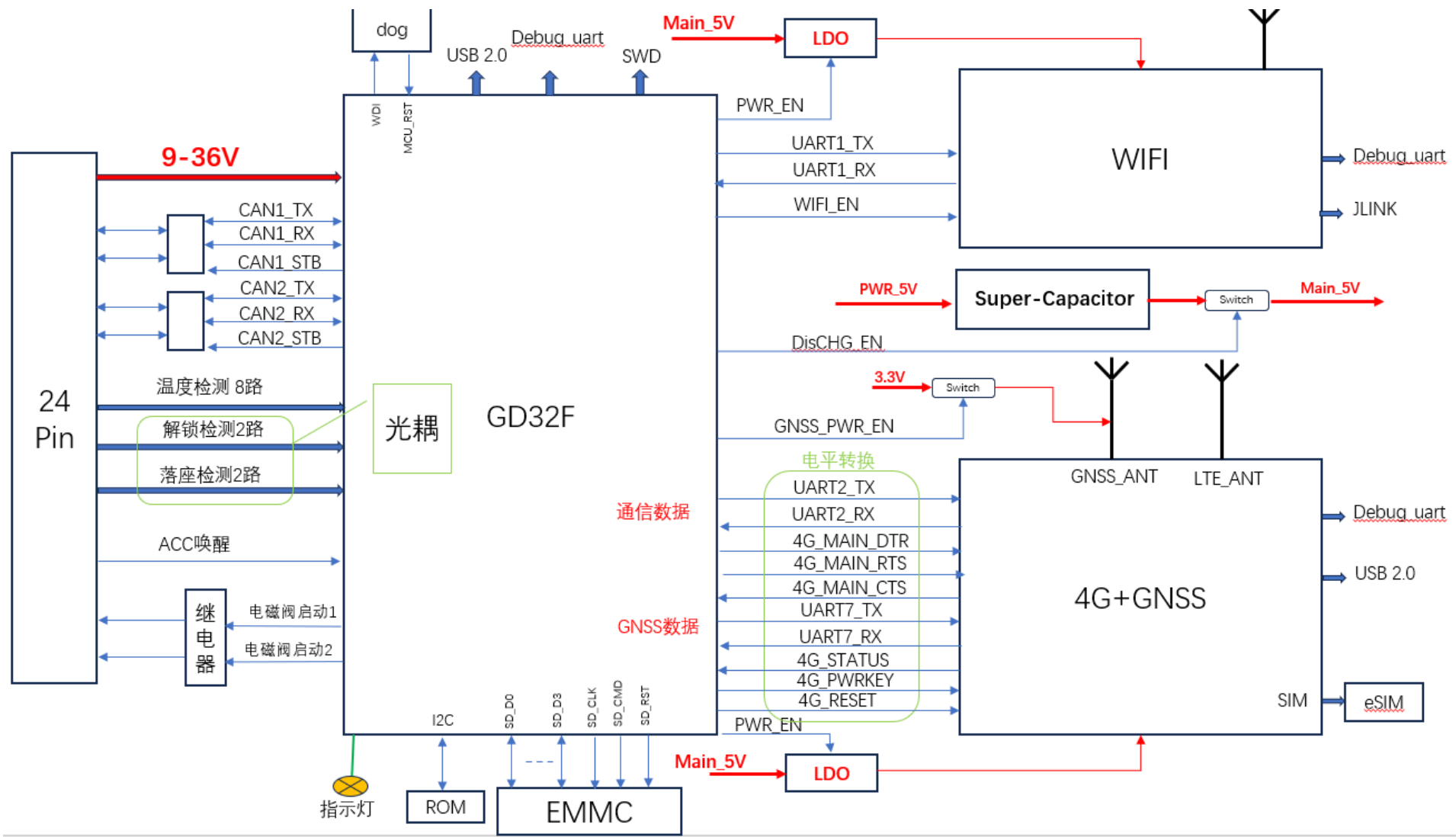


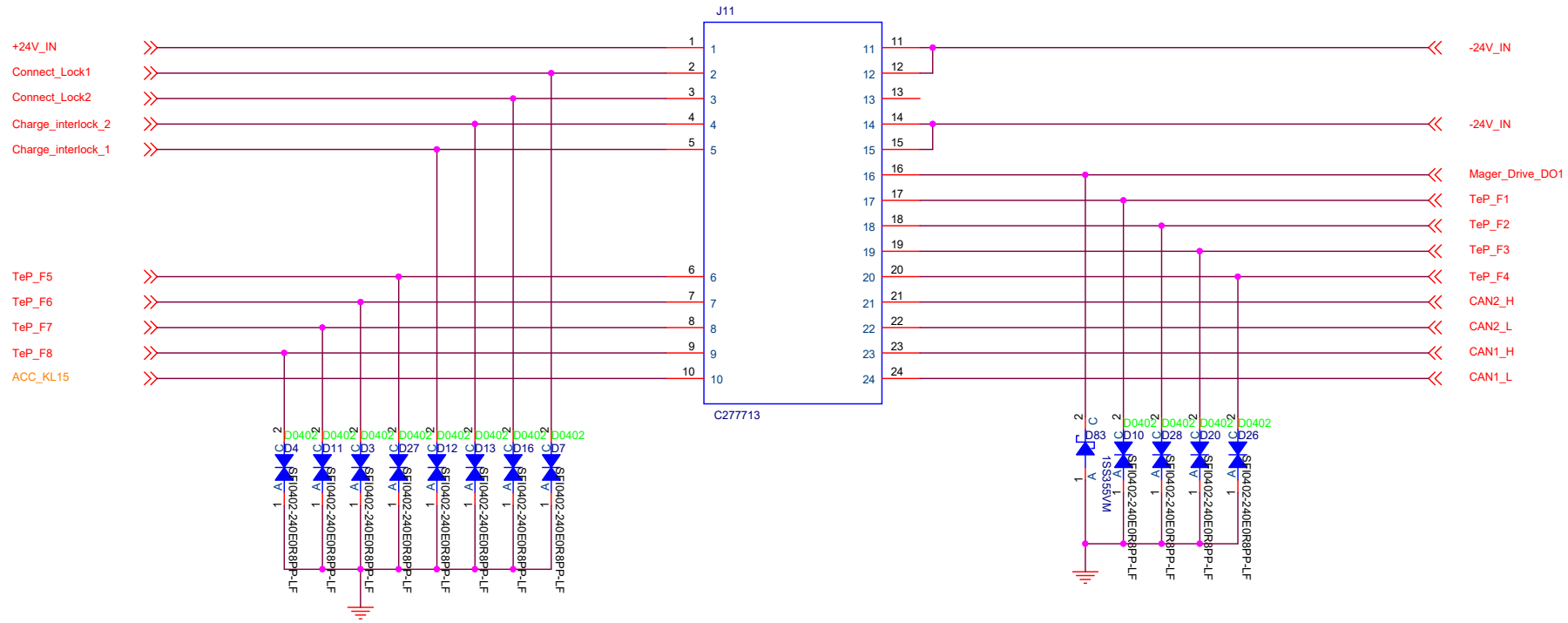
SPIC_TB0X1.2_H01

Revision History

Revision	Date	Comments
A	03/10/2025	Initial release

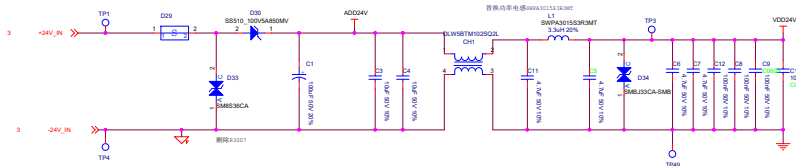
	
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Page Contents: Title	Rev: VA01
Author: <WZL>	Size: B
Date: Wednesday, April 02, 2025	Sheet 1 of 14



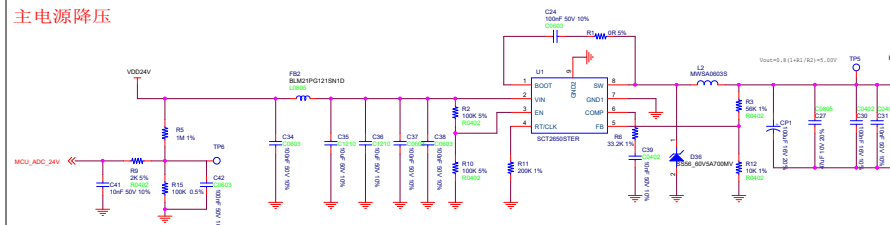


 启源芯动力 SPIC QIYUAN XIN DONGLI	
Title: SPIC_TBOX2.2_H01	
Page Contents: Interface	Rev: <H01>
Author: <WZL>	Size: B
Date: Wednesday, April 02, 2025	Sheet 3 of 14

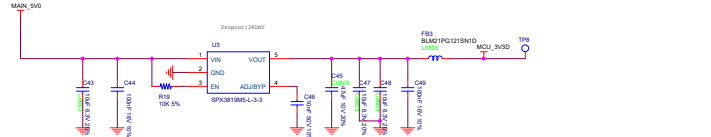
车载输入电源滤波



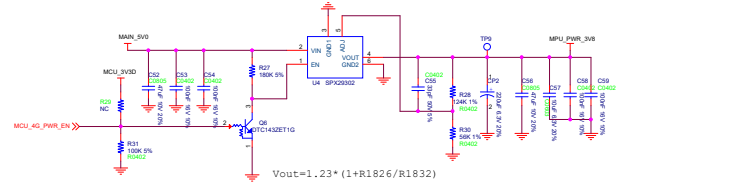
主电源降压



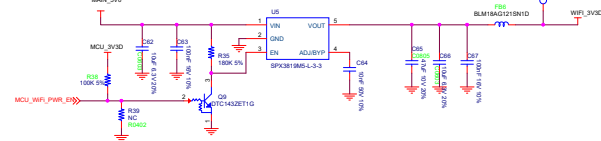
MCU供电



4G模组供电



WIFI供电电路



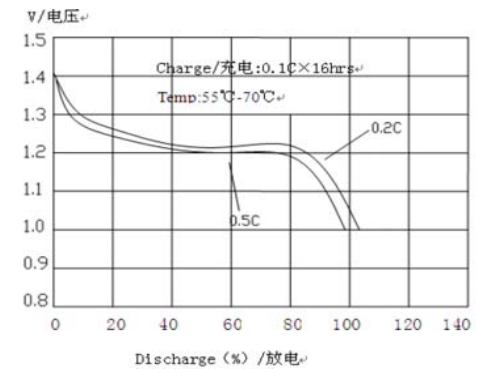
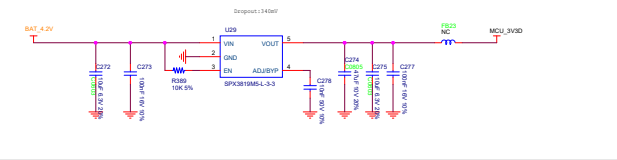
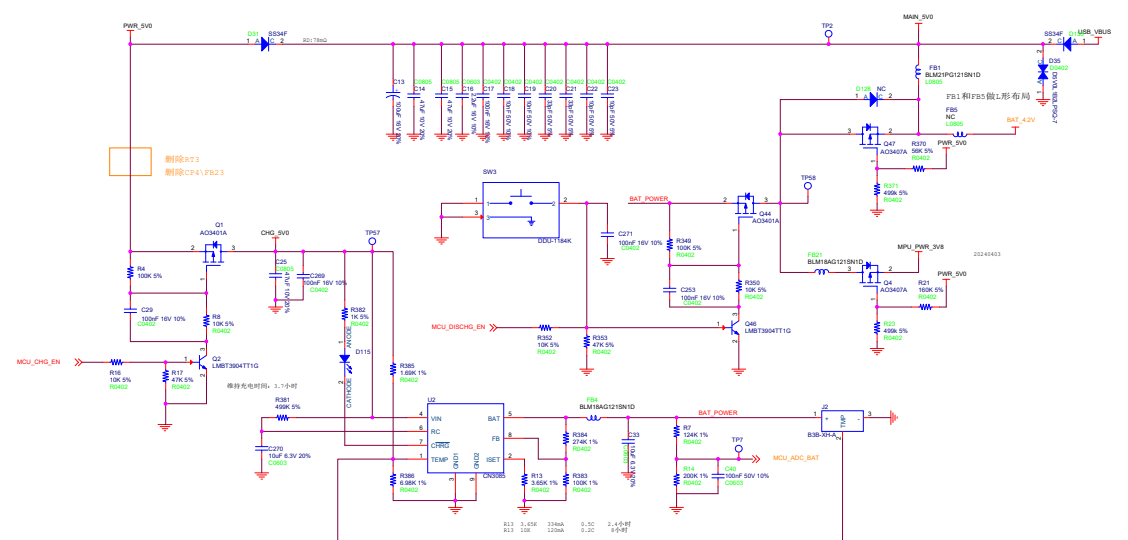
车载电源掉电中断

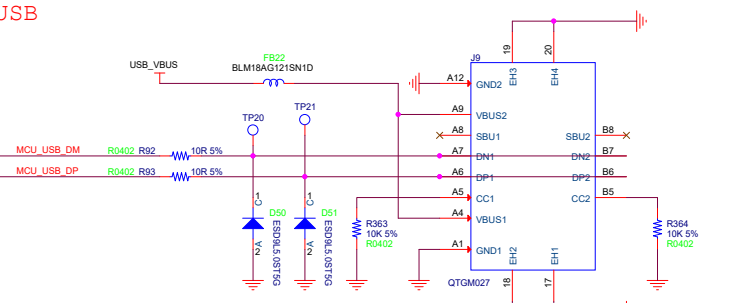
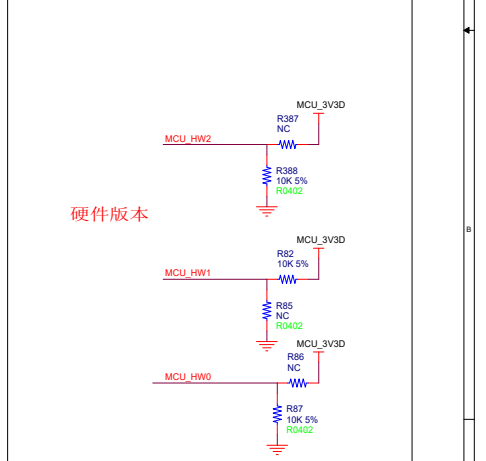
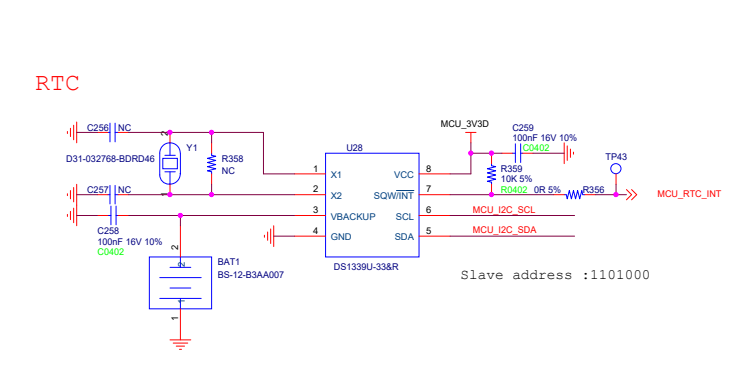
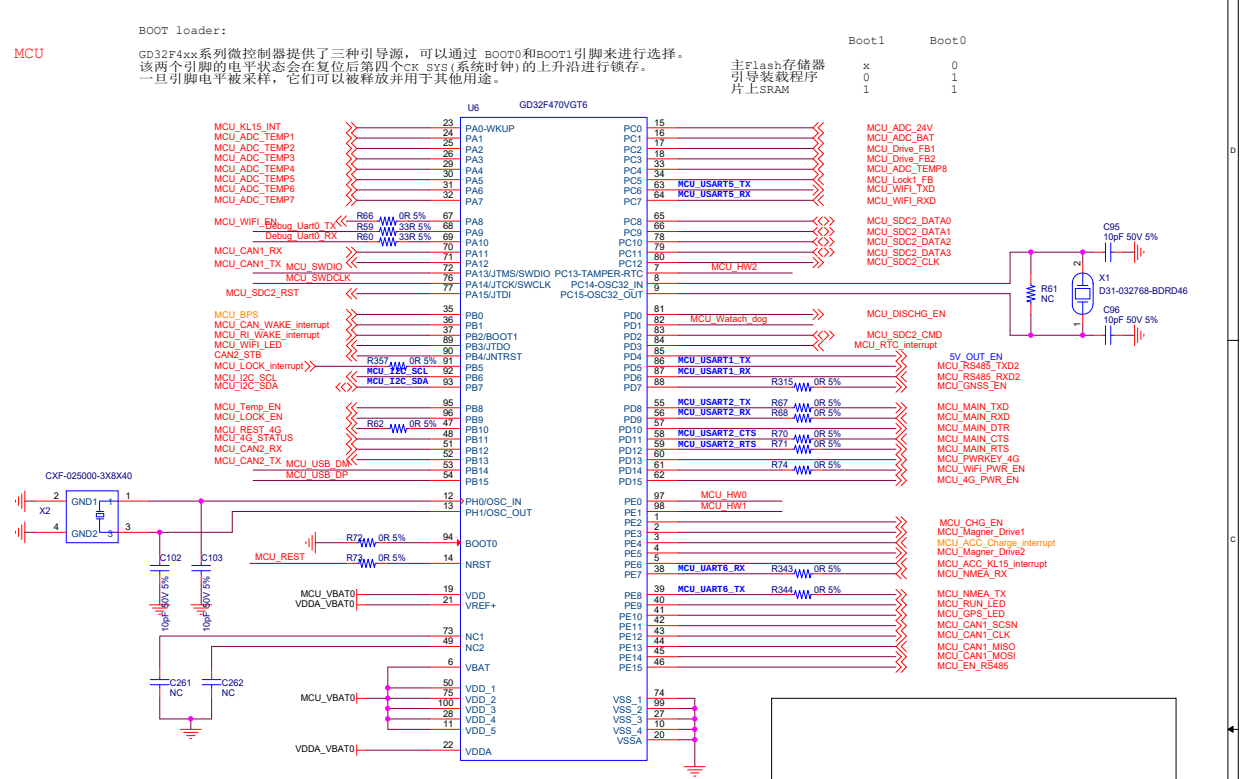
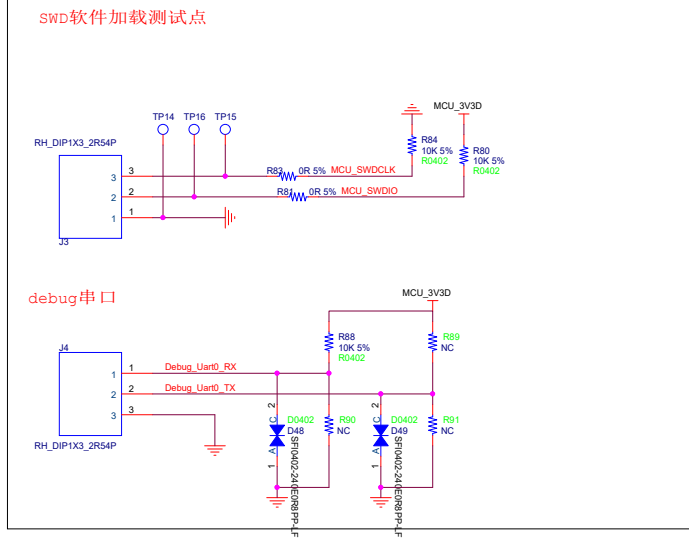
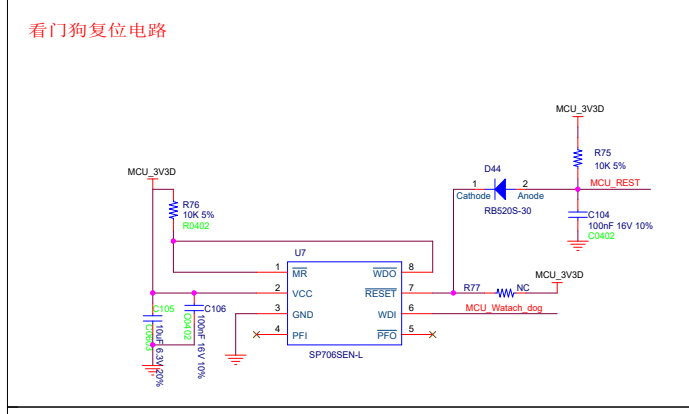
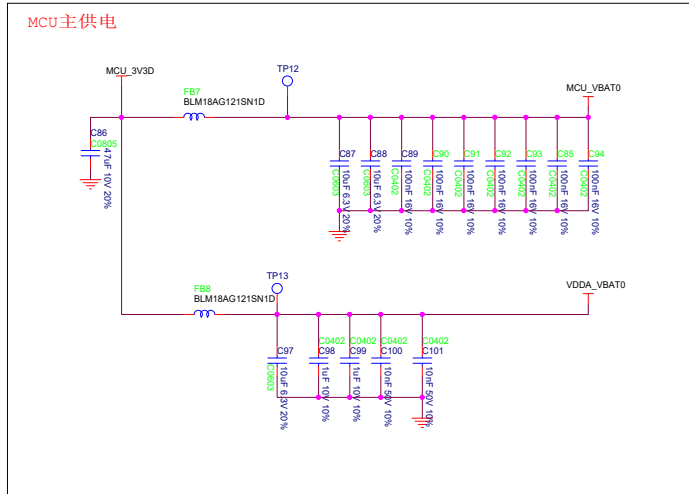
20240417-改

5V电源输出开关

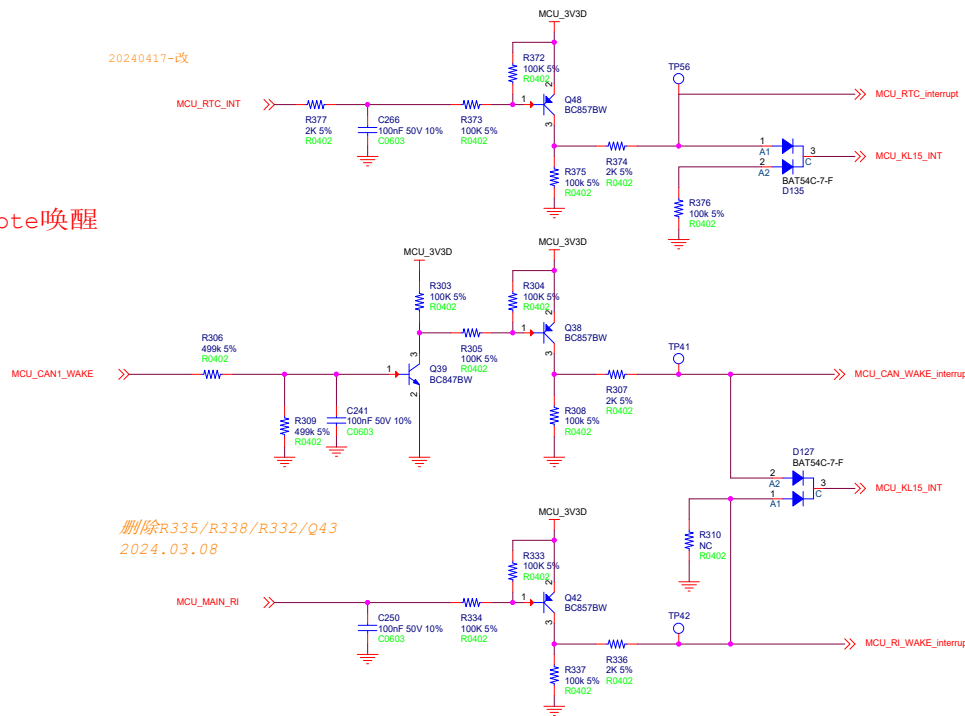


超级电容供电

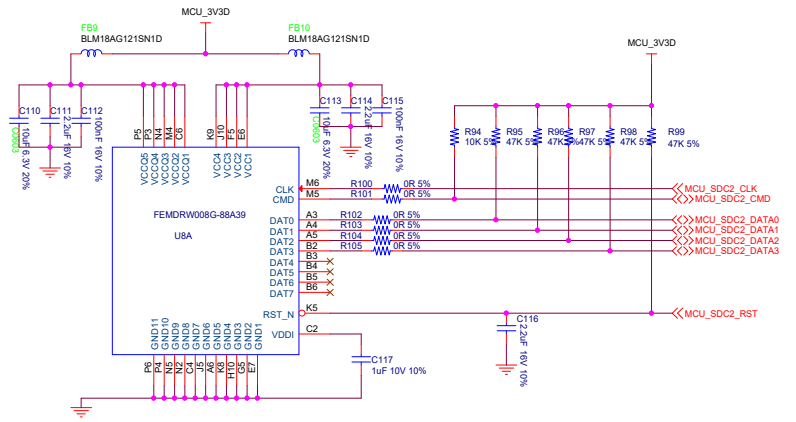




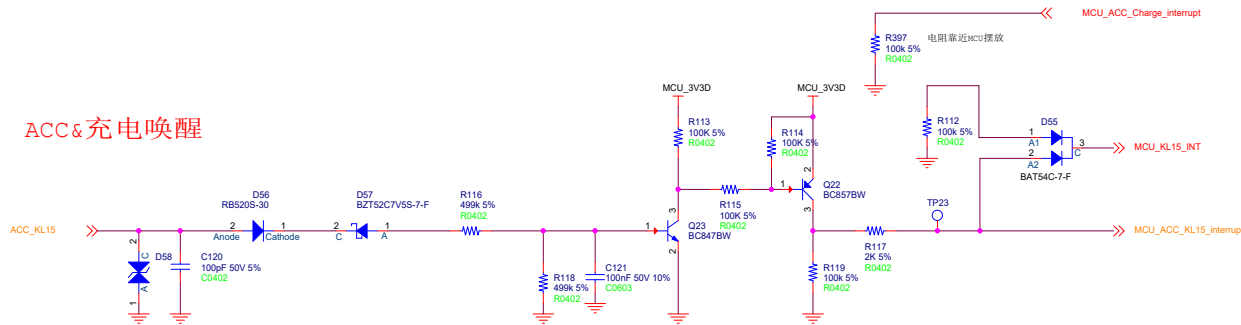
CAN&Remote唤醒



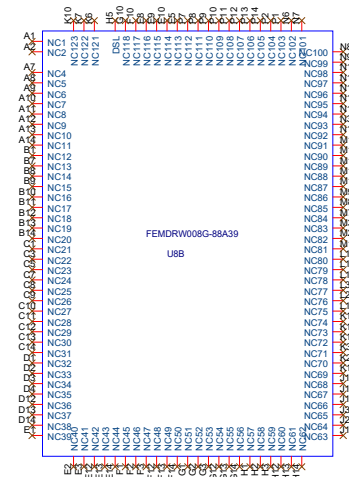
EMMC电路



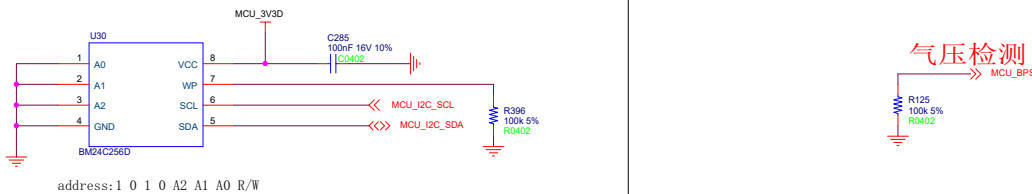
ACC&充电唤醒



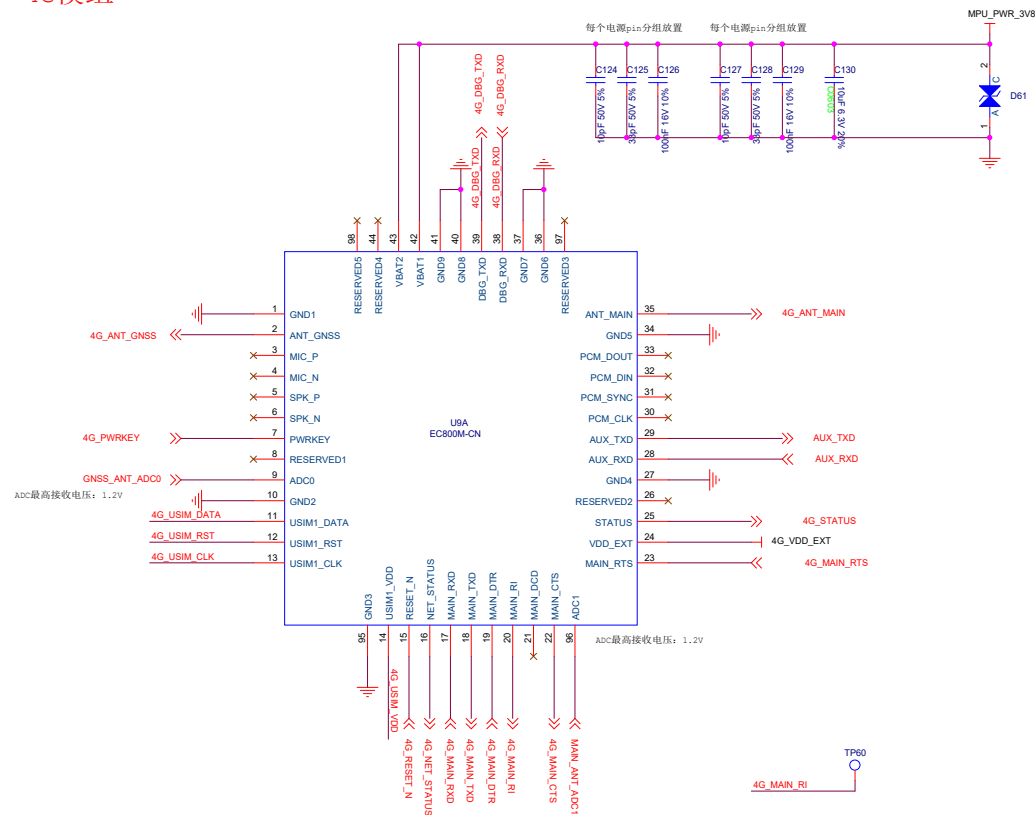
EMMC电路



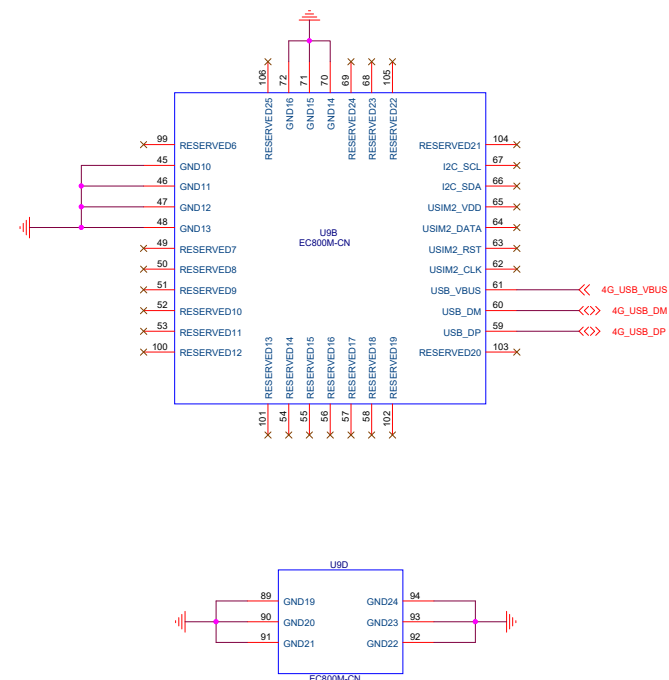
气压检测



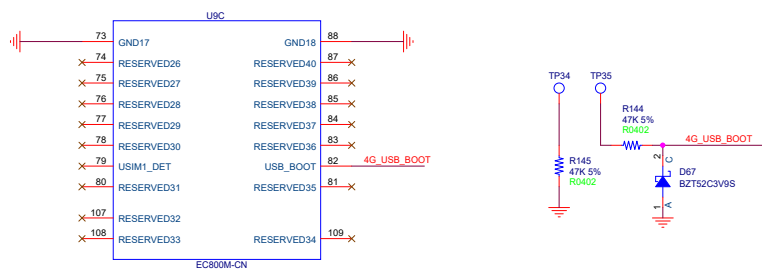
4G模组



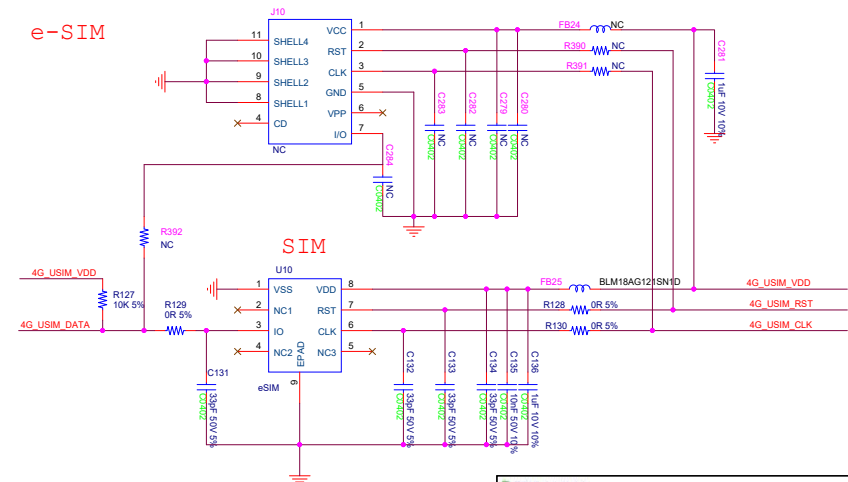
4G模组



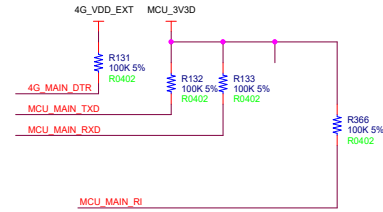
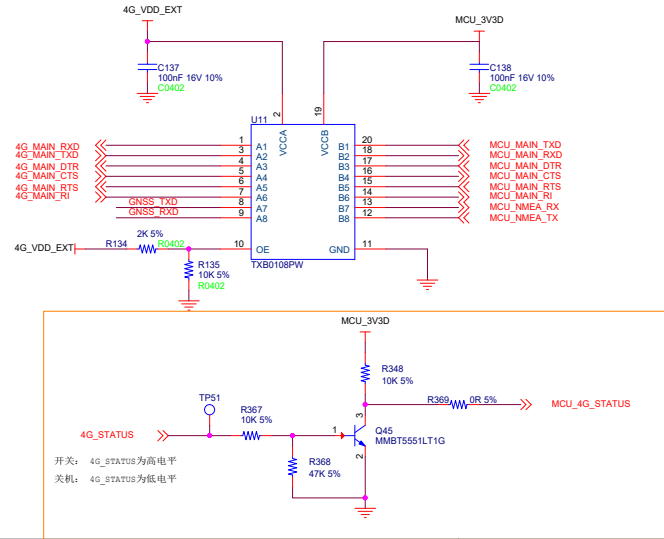
4G模组



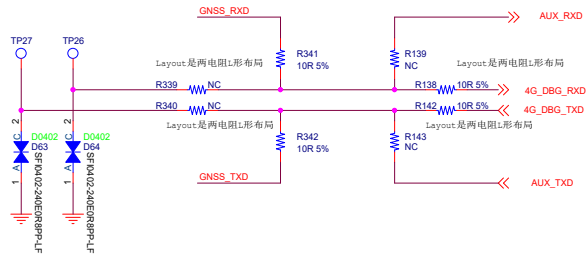
e-SIM



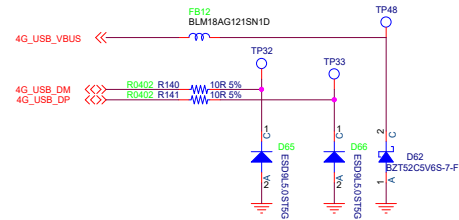
电平转换



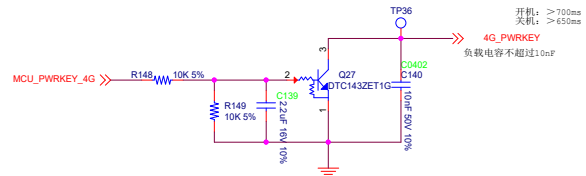
4G模组debug串口



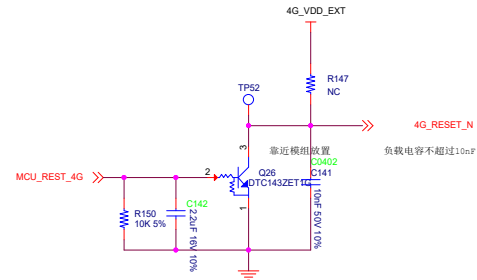
4G模组USB口



4G模组开/关机



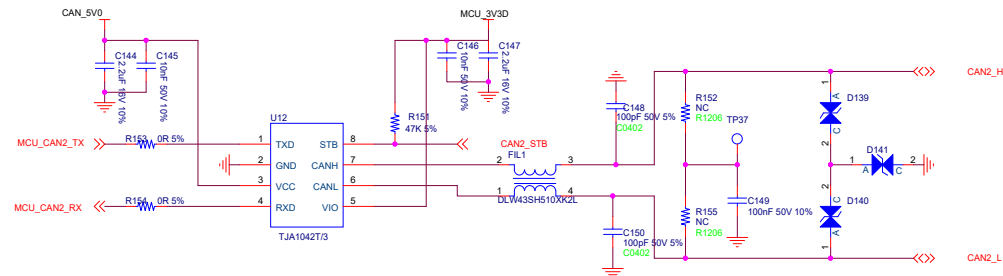
4G模组复位



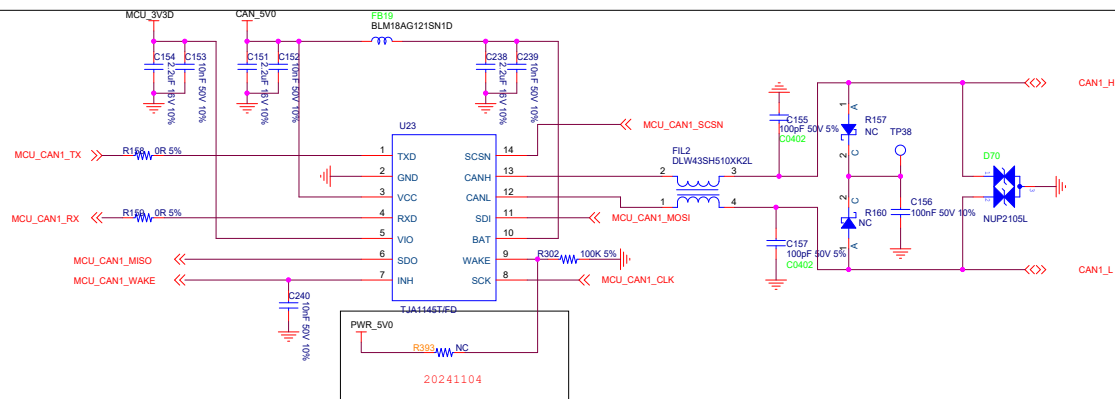
CAN电源



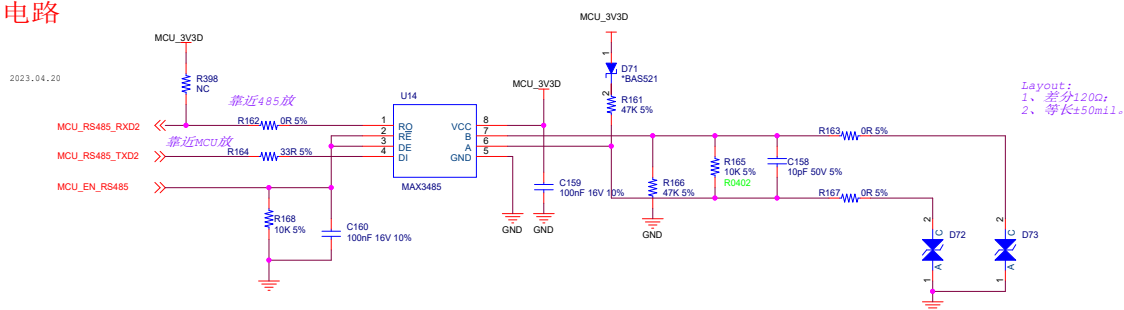
CAN2

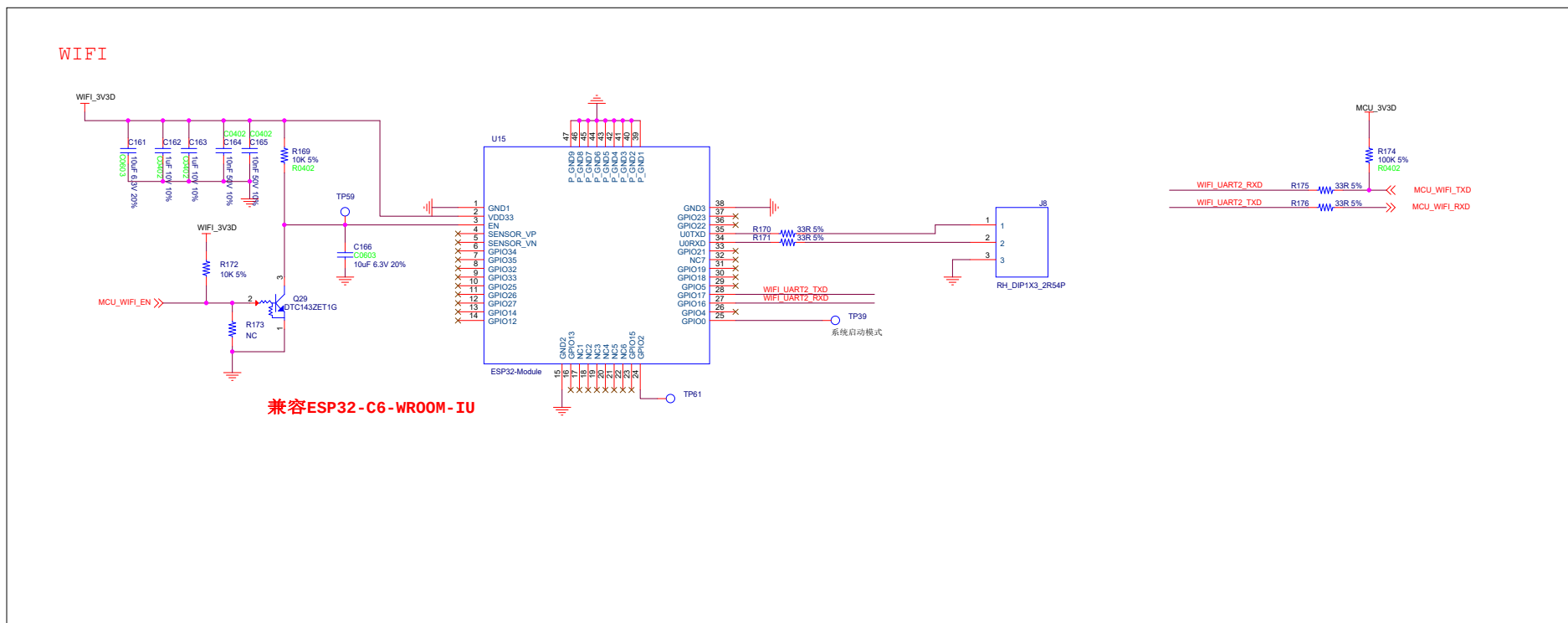


CAN1



RS485电路





2.4 GHz Wi-Fi 6 (802.11ax), Bluetooth® 5 (LE), Zigbee 及 Thread (802.15.4) 模组
 内置 ESP32-C6 系列芯片，RISC-V 32 位单核处理器
 flash 最大可选 8 MB
 23 个 GPIO，丰富的外设
 板载 PCB 天线或外部天线连接器



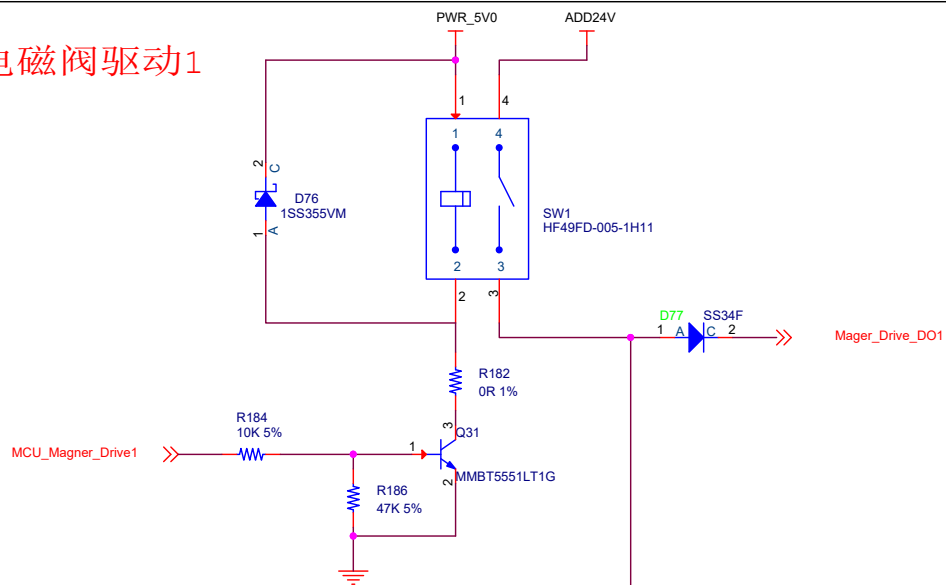
ESP32-C6-WROOM-1



ESP32-C6-WROOM-1U

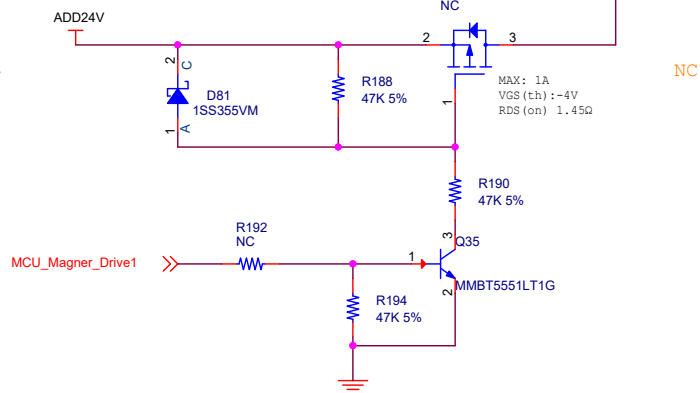
管脚	默认	3.3 V	1.8 V
MTDO	下拉	0	1
管脚	默认	SPI 启动模式	下载启动模式
GPIO0	上拉	1	0
GPIO2	下拉	无关联	0
管脚	默认	系统启动过程中，控制 UOTXD 打印	
MTDO	上拉	UOTXD 正在打印	UOTXD 上电不打印
管脚	默认	1	0
管脚	默认	SDIO 从机信号输入输出时序	
MTDO	上拉	下降沿采样 下降沿输出	上升沿采样 下降沿输出
GPIO5	上拉	0	1

电磁阀驱动1

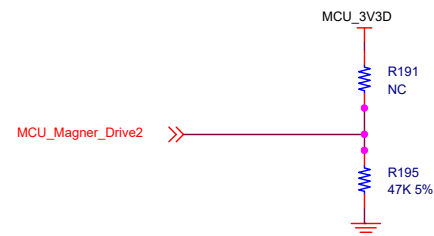


电磁阀驱动2

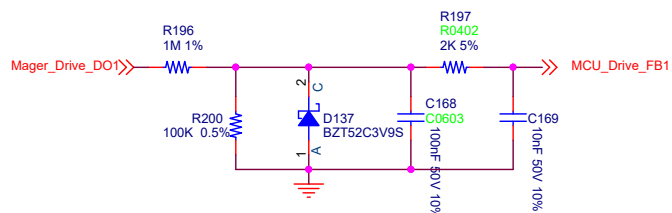
OR



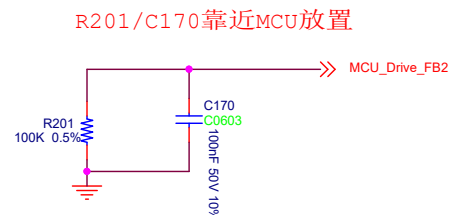
OR



电磁阀驱动FB1



电磁阀驱动FB2



Title: SPIC_TBOX2.2_H01

Page Contents: Power_Relay

Author: <WZL>

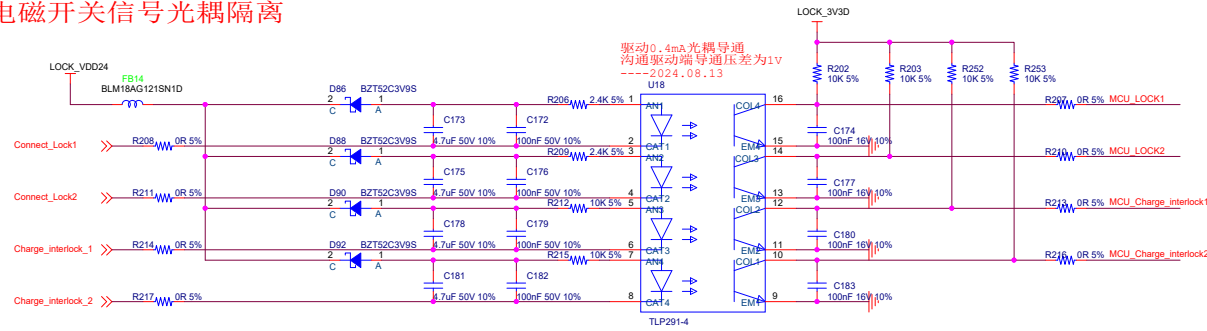
Date: Wednesday, April 02, 2025

Rev: <H01>

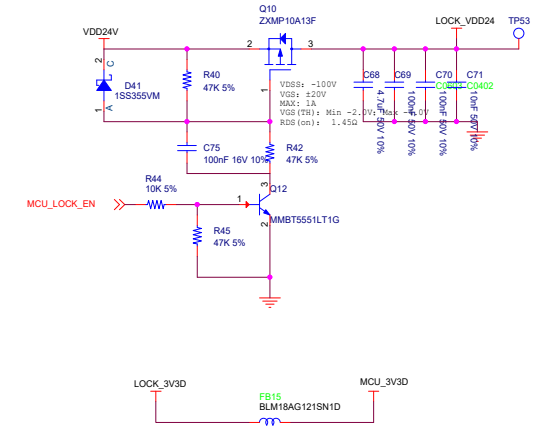
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Sheet 11 of 14

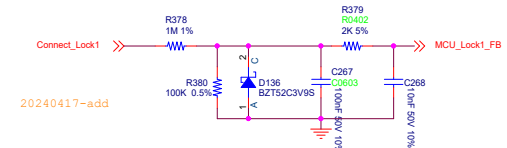
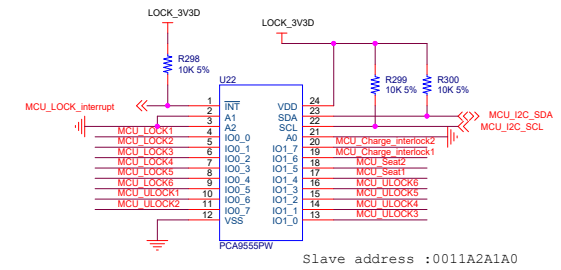
电磁开关信号光耦隔离



光耦测试电源开关



底托磁开信号采集



Title: SPIC_TB0X2.2_H01

Page Contents: Lock_Ulock_Monitor

Author: <WZL>

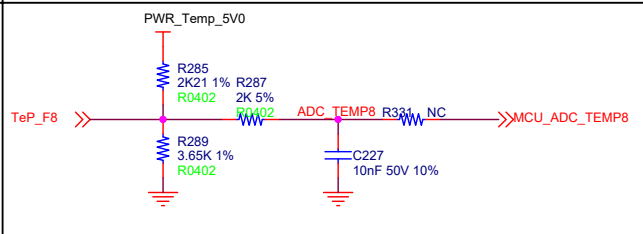
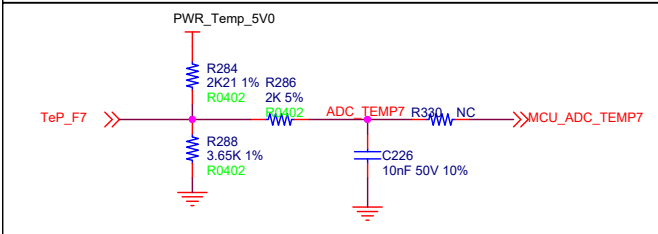
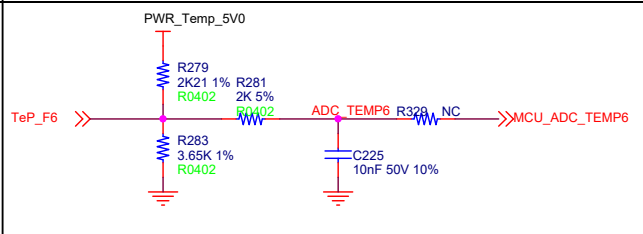
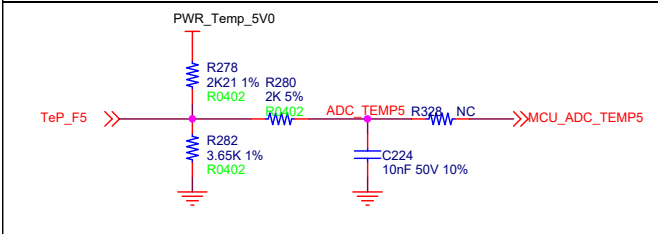
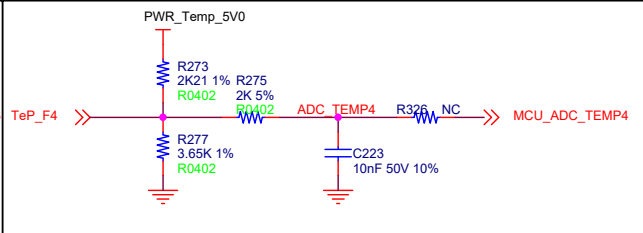
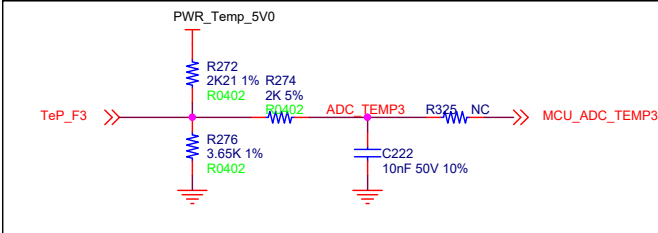
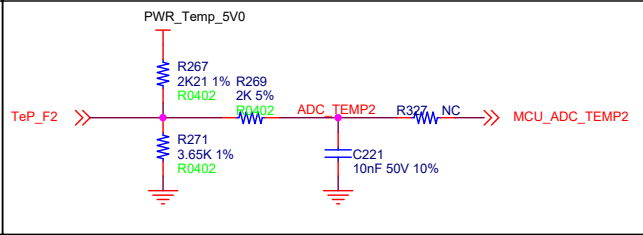
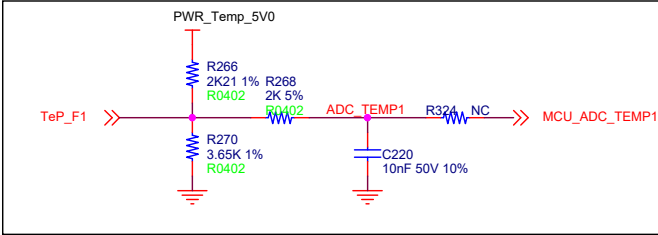
Date: Wednesday, April 02, 2025

Rev: <H01>

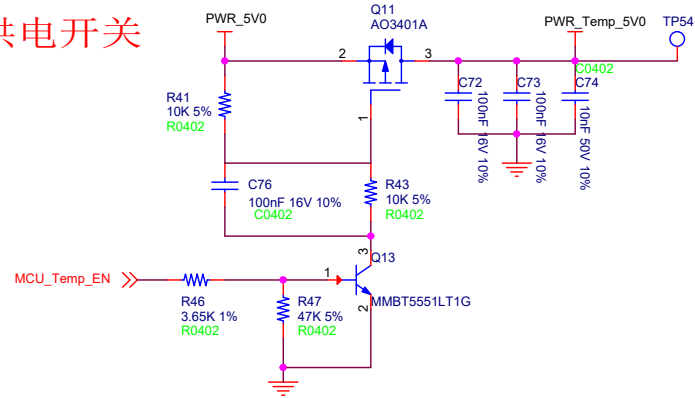
Size: C

Sheet 12 of 14

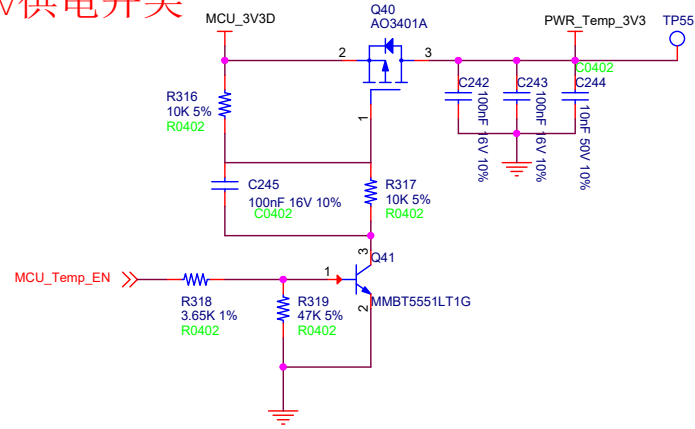
温度检测



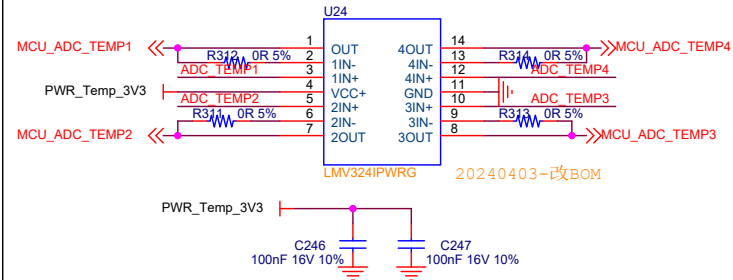
5v供电开关



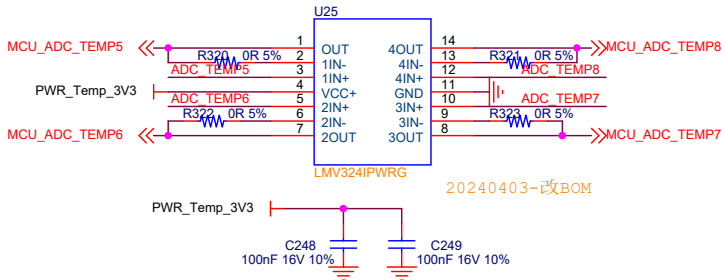
3.3v供电开关



跟随器隔离

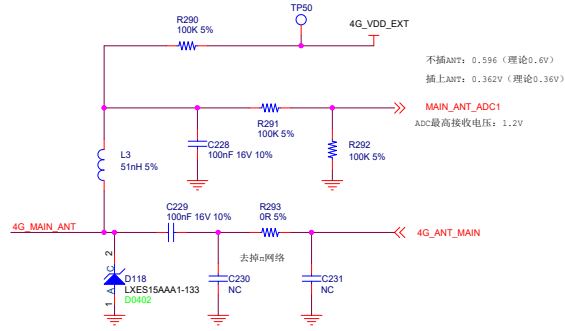


跟随器隔离

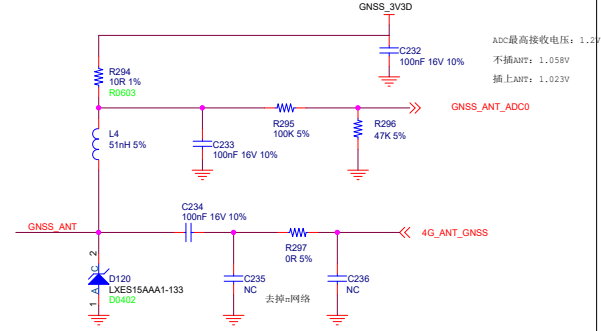


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Page Contents: Temp_Monitor	Rev: <H01>
Author: <WZL>	Size: B
Date: Wednesday, April 02, 2025	Sheet 13 of 14

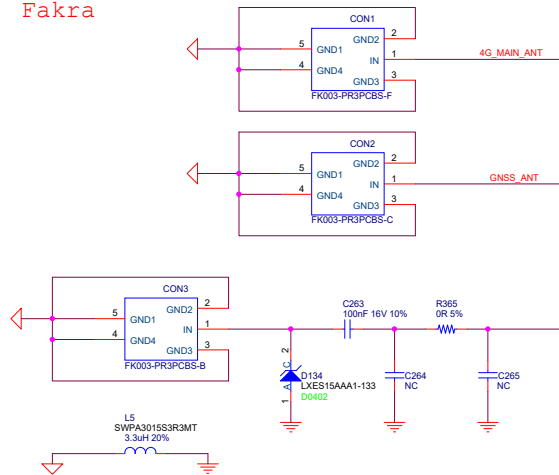
4G MAIN ANT



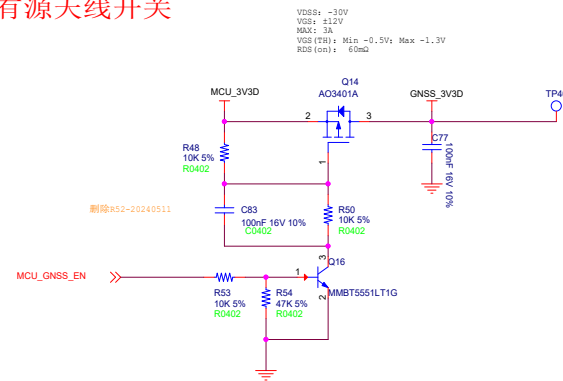
GNSS ANT



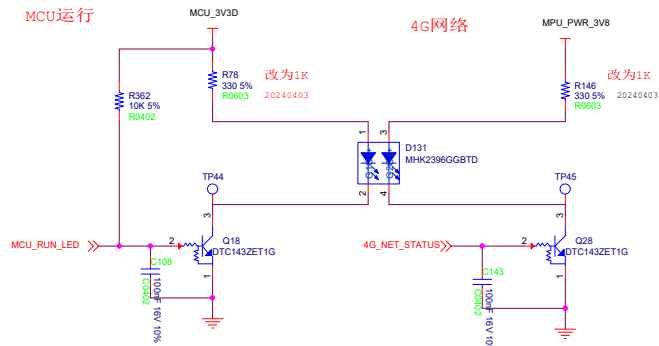
Fakra



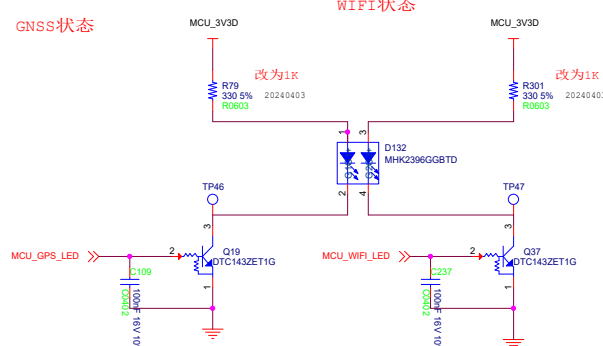
GNSS有源天线开关



MCU运行



GNSS状态



WIFI状态



Title: SPIC_TBOX2.2_H01

Page Contents: ANT

Author: <WZL>

Date: Wednesday, April 02, 2025

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Size: C

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